
Music Context Understanding with DistilBERT and Segment-Graph Fusion

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Abstract

Music can be understood through different types of information, such as genre, instruments, vocals, and descriptive tags. In this project, we study music context using both text and audio information. We conduct three main experiments. First, we use DistilBERT to predict music tags from MusicCaps captions. Second, we use GraphSAGE and a CNN for audio-based genre classification on the GTZAN dataset. Third, we combine GraphSAGE and DistilBERT for multi-label music tagging on MagnaTagATune (MTAT). The MusicCaps DistilBERT model achieves a Micro-F1 of 0.4817 and a Macro-F1 of 0.2775. On GTZAN, the CNN performs better than GraphSAGE, achieving a Macro-F1 of 0.7112 compared with 0.3258. On MTAT, early concatenation gives the best fusion result, with 0.3654 Macro-F1, 0.3937 Micro-F1, and 0.3294 mAP. Overall, the results show that combining audio and text information can improve music-context prediction. However, simple audio features and overlapping music tags remain important limitations.

1 Introduction

Music can contain different types of information, such as genre, instruments, vocals, and other descriptive characteristics. A single music clip may also belong to several categories at the same time. For example, a song can be described as rock, fast, guitar-based, and male vocal. Therefore, understanding music context often requires predicting multiple labels rather than only one class.

In this project, we study how text and audio information can be used to understand music context. For text, we use DistilBERT to learn information from music captions and user tags. For audio, we divide each music clip into several segments and represent them as a graph. GraphSAGE is then used to learn relationships between nearby and acoustically similar audio segments. We also use a CNN on mel-spectrograms as an audio baseline for comparison.

Our experiments are performed on three music datasets. First, we use MusicCaps to predict multiple music tags from natural-language captions. Next, we use GTZAN to compare graph-based genre classification with a conventional CNN-based approach. Finally, we use MagnaTagATune (MTAT) to combine audio-graph features with text features for multi-label music tagging. We compare BERT-only and GNN-only models with early concatenation and cross-attention fusion. We also evaluate a partially end-to-end trained cross-attention model.

These experiments use different datasets and different label sets, so their scores should not be directly compared with each other. Instead, each experiment helps us understand a different part of the problem. Our results show that the CNN performs much better than the chroma-based GraphSAGE model for GTZAN genre classification. However, on MTAT, combining the graph and text representations improves the results compared with using either representation alone. Among the tested fusion methods, early concatenation gives the best overall performance.

2 Related Work

DistilBERT is a smaller and faster version of BERT that can still produce useful text representations [1]. In this project, we use DistilBERT to understand music-related text, including captions and user tags. The text representation is then used to predict music-context labels. We use DistilBERT as a pretrained text encoder rather than studying the model compression process itself.

GraphSAGE learns information about a node by combining its own features with information from neighboring nodes [2]. We use this idea to represent the structure of music clips. Each clip is divided into six audio segments, where each segment becomes a graph node. Connections are created between neighboring segments and between segments with similar chroma features. GraphSAGE then combines information from these connected segments to create one representation of the music clip.

Tzanetakis and Cook [3] showed that audio features such as timbre, rhythm, and pitch can be useful for music genre classification. Following this idea, we compare our chroma-based GraphSAGE model with a CNN that uses mel-spectrograms on the GTZAN dataset.

MusicCaps contains expert-written text descriptions of music clips and was introduced with MusicLM [4]. We use these descriptions for music-tag prediction rather than music generation. For multi-label music tagging, we also use MagnaTagATune (MTAT). Previous work by Law et al. [5] shows that music tags can describe different properties of a song and that available tag annotations may not always be complete. This is important when interpreting the predictions of our MTAT models.

Overall, these studies provide the main ideas behind our use of DistilBERT, GraphSAGE, music datasets, and audio classification methods. Our project builds on these ideas by studying how text and graph-based audio information can be combined for music-context prediction.

3 Method

3.1 Tasks and Datasets

We use three datasets for three separate prediction problems. Table 1 summarizes the processed splits. Each dataset has its own labels and model, so results across the three tasks are not directly comparable.

Table 1: Processed datasets and fixed train, validation, and test splits.

Task / dataset	Model input	Train	Valid.	Test	Targets
1 / MusicCaps	Caption text	3,996	499	500	50 tags
2 / GTZAN	Audio only	799	100	100	10 genres
3 / MTAT	Audio + user tags	18,706	1,825	5,329	50 tags

Caption-to-tag prediction: MusicCaps contains 5,521 captioned examples. We select the 50 most frequent aspect tags and keep 4,995 examples containing at least one selected tag. Captions are used as input and the selected tags as multi-label targets. The tag vocabulary is chosen before the split, so overall label frequency contributes to task definition. This experiment evaluates text-based music understanding.

Audio-based genre classification: GTZAN contains ten music genres [3]. One unreadable file, `jazz.00054.wav`, is removed, leaving 999 tracks. GraphSAGE and the CNN use the same stratified split of 799 training, 100 validation, and 100 test tracks.

Multimodal music tagging: For MTAT, we use the standard `musicnn-training` benchmark split with 18,706 training, 1,825 validation, and 5,329 test examples. From the original 188 annotations, 50 tags are used as prediction targets and the remaining 138 tags form the auxiliary text input. Exact target columns are excluded, although related words such as “violins” and “violin” may still occur. We therefore describe this setting as *metadata-assisted music tagging*.

3.2 Audio-Segment Graphs

For the graph-based models, each audio clip is converted to mono at a sampling rate of 22,050 Hz. Each clip is padded or trimmed to 30 seconds and divided into six non-overlapping segments of five seconds each.

For every segment, we calculate a 12-dimensional chroma vector. Chroma represents the strength of the 12 musical pitch classes. Each audio segment becomes one node in the graph.

Two types of connections are used. First, consecutive segments are always connected because they occur next to each other in time. Second, two non-adjacent segments are connected when their chroma features are highly similar. We measure this using cosine similarity with a threshold of 0.99:

$$(i, j) \in E \iff |i - j| = 1 \text{ or } \left(|i - j| > 1 \text{ and } \frac{c_i^\top c_j}{\|c_i\| \|c_j\|} \geq 0.99 \right). \quad (1)$$

Here, c_i and c_j represent the chroma features of two audio segments. In simple terms, the graph connects segments that are close in time or acoustically similar. Both directions of every connection are stored for GraphSAGE message passing.

For MTAT, the similarity threshold of 0.99 was checked using 2,000 non-adjacent segment pairs from 200 training clips. About 13.15% of these pairs reached this similarity level, so the threshold was fixed at 0.99 for the remaining experiments.

For GTZAN, the original chroma features are used as node features. For MTAT, the chroma features are standardized within each music clip before being passed to GraphSAGE. However, graph connections are still created using the original chroma values.

We use two GraphSAGE layers. Each layer updates a node by combining its own features with the average features of its connected neighbors:

$$h_i^{(\ell+1)} = \text{ReLU} \left(W_s^{(\ell)} h_i^{(\ell)} + W_n^{(\ell)} \text{mean}_{j \in \mathcal{N}(i)} h_j^{(\ell)} + b^{(\ell)} \right). \quad (2)$$

After the second GraphSAGE layer, mean pooling combines the six node representations into one representation of the full music clip:

$$g = \frac{1}{6} \sum_{i=1}^6 h_i^{(2)}. \quad (3)$$

The hidden size is 64 for GTZAN and 128 for MTAT. Dropout of 0.3 is applied after the first GraphSAGE layer. The final graph representation is used for GTZAN genre classification and for MTAT music-tag prediction and fusion.

These graphs represent relationships between different audio segments. They do not directly detect or assign musical chord names.

3.3 Text Encoding and CNN Baseline

DistilBERT is used to encode MusicCaps captions and MTAT auxiliary tags. Text is tokenized using the uncased DistilBERT tokenizer and limited to 128 tokens. MusicCaps uses a 50-label classification head. For MTAT, DistilBERT provides a 768-dimensional text representation for BERT-only and early-concatenation models, while cross-attention uses all token representations. Dropout of 0.2 is applied before prediction. When auxiliary MTAT tags are missing, we use the fixed phrase *music context: no auxiliary tags*.

For GTZAN, we use a CNN as an audio-only baseline. Each track is converted to a 128-bin log-mel spectrogram using an FFT size of 1,024 and hop length of 512, then standardized and resized to 128×256 . The CNN has three convolutional blocks with 16, 32, and 64 channels, followed by adaptive 4×4 pooling, a 128-unit dense layer, dropout 0.3, and a 10-class output.

The models are implemented with PyTorch, PyTorch Geometric, and Hugging Face Transformers. MTAT experiments were run on Kaggle using a Tesla T4 GPU.

3.4 Multimodal Fusion and Training Objectives

The MTAT experiment combines the graph-based audio representation with the DistilBERT text representation. Figure 1 shows the complete multimodal pipeline.

The audio branch uses the segment graph described in Section 3.2. GraphSAGE processes the six audio nodes and produces a 128-dimensional graph representation g . The text branch processes the auxiliary user tags using DistilBERT and produces a 768-dimensional text representation t together with token-level representations.

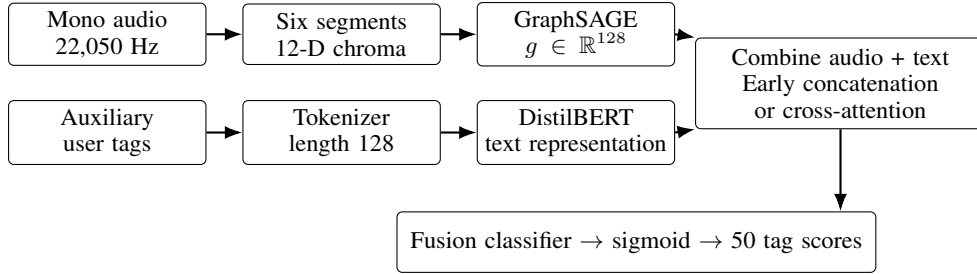


Figure 1: Overview of the multimodal music-tagging model. The audio clip is divided into six segments and processed by GraphSAGE, while the auxiliary user tags are processed by DistilBERT. The audio and text representations are then combined using early concatenation or cross-attention to predict 50 music tags.

We compare two fusion methods: early concatenation and cross-attention.

In early concatenation, the 128-dimensional graph representation g is joined with the 768-dimensional DistilBERT representation t . This creates an 896-dimensional vector:

$$u = [g; t] \in \mathbb{R}^{896}. \quad (4)$$

The combined vector is passed through layer normalization, a 256-unit ReLU layer, dropout of 0.3, and a final prediction layer for the 50 MTAT tags.

Cross-attention combines the two sources differently. The graph representation is first projected into the same 768-dimensional space as the DistilBERT token representations. The projected graph representation is used as the query, while the DistilBERT token representations are used as the keys and values. We use eight attention heads:

$$a = \text{softmax} \left(\frac{qK^\top}{\sqrt{d_h}} + M \right) V, \quad d_h = 96. \quad (5)$$

Here, q is the projected graph representation, K and V are the DistilBERT token representations, and M masks padded tokens. Since the text representation has 768 dimensions and eight attention heads are used, each head has 96 dimensions.

The attended text representation is then combined with the projected graph representation. The combined vector is passed through layer normalization, a 256-unit ReLU layer, dropout, and a prediction layer for the 50 music tags. Attention dropout is 0.1 and classifier dropout is 0.3.

Because MTAT is a multi-label problem, one music clip may contain several correct tags at the same time. We therefore use weighted binary cross-entropy so that less frequent positive tags receive more importance:

$$\mathcal{L}_{\text{tags}} = -\frac{1}{BK} \sum_{n=1}^B \sum_{k=1}^K [w_k y_{nk} \log p_{nk} + (1 - y_{nk}) \log(1 - p_{nk})]. \quad (6)$$

The weight for each tag is calculated as

$$w_k = \text{clip} \left(\frac{N - n_k^+}{n_k^+ + \epsilon}, 1, 20 \right), \quad \epsilon = 10^{-6}. \quad (7)$$

Here, B is the batch size, K is the number of tags, N is the number of training examples, and n_k^+ is the number of positive training examples for tag k .

For the MusicCaps multi-label experiment, we use standard binary cross-entropy without these class weights. For GTZAN genre classification, we use multiclass cross-entropy because each track has only one genre label.

We do not perform valence or arousal prediction because our experiments do not contain continuous emotion labels.

4 Experiments

4.1 Training Protocol

We use Adam to train the GraphSAGE and CNN models, while DistilBERT and the fusion models are trained using AdamW. Table 2 summarizes the main training settings.

For MusicCaps, the best DistilBERT checkpoint is selected using validation Micro-F1. For GTZAN, the best GraphSAGE and CNN checkpoints are selected using validation Macro-F1. For MTAT, the best checkpoints are selected mainly using validation mean average precision (mAP). In the frozen fusion models, the pretrained GraphSAGE and DistilBERT encoders are kept fixed and only the fusion layers are trained.

Table 2: Main training settings used in the experiments. Max/best shows the maximum number of training epochs and the epoch selected as the best checkpoint.

Model / task	Batch	Learning rate	Weight decay	Max/best
DistilBERT / MusicCaps	16	2×10^{-5}	10^{-2}	3/3
GraphSAGE / GTZAN	32	10^{-3}	10^{-4}	50/27
CNN / GTZAN	32	10^{-3}	10^{-4}	30/24
DistilBERT / MTAT	16	2×10^{-5}	10^{-2}	6/4
GraphSAGE / MTAT	128	10^{-3}	10^{-4}	50/28
Early concat / MTAT	256	10^{-3}	10^{-4}	15/11
Frozen attention / MTAT	64	5×10^{-4}	10^{-4}	6/5
Partial end-to-end / MTAT	32	Separate rates*	10^{-4}	2/2

*For partial end-to-end training, the learning rates are 5×10^{-5} for GraphSAGE, 10^{-5} for the final DistilBERT block, and 2×10^{-4} for the fusion layers.

For the partial end-to-end model, the GraphSAGE encoder, the final DistilBERT block, and the fusion layers are updated together. Gradient clipping with a maximum norm of 1 is used to make training more stable. The model starts from the previously trained frozen cross-attention model.

Early stopping is also used for the frozen fusion models. Early concatenation uses a patience of three epochs, while frozen cross-attention uses a patience of two epochs. The different models do not use exactly the same number of training epochs because their validation performance stops improving at different times.

4.2 Evaluation and Reproducibility

We report Macro-F1 and Micro-F1 for multi-label prediction, and accuracy for GTZAN. For MTAT, we also report mean average precision (mAP), calculated by averaging `average_precision_score` across the 50 tags.

Decision thresholds are selected using only the validation set. MusicCaps tests thresholds from 0.05 to 0.50 and selects the value with the highest Micro-F1. MTAT tests thresholds from 0.05 to 0.95 and selects the value with the highest Macro-F1. The selected values are then applied to the test set.

Seed 42 is used for the data splits and final fusion experiments. We do not report confidence intervals because the experiments were not repeated across multiple seeds. MTAT clip IDs are disjoint across splits, although artist-disjointness was not independently verified.

5 Results

5.1 Task 1: Caption-to-Tag Prediction

The MusicCaps DistilBERT model achieves a Micro-F1 of 0.4817 and a Macro-F1 of 0.2775 on the 500 test captions. The model also obtains a micro-precision of 0.4676 and a micro-recall of 0.4968.

These results measure how well the model predicts selected music tags from their associated text captions. Therefore, this experiment evaluates text-based music understanding rather than prediction from audio.

During training, the validation loss decreases from 0.2035 in the first epoch to 0.1654 in the third epoch. At the default threshold of 0.5, validation Micro-F1 remains relatively low. We therefore test several decision thresholds using only the validation set.

Figure 2 shows that both validation Micro-F1 and Macro-F1 increase across the three training epochs when evaluated at the default threshold of 0.5. After training, we separately tune the decision threshold using the validation set. A threshold of 0.20 gives the highest validation Micro-F1 of 0.4839 and is therefore used for the final test evaluation.

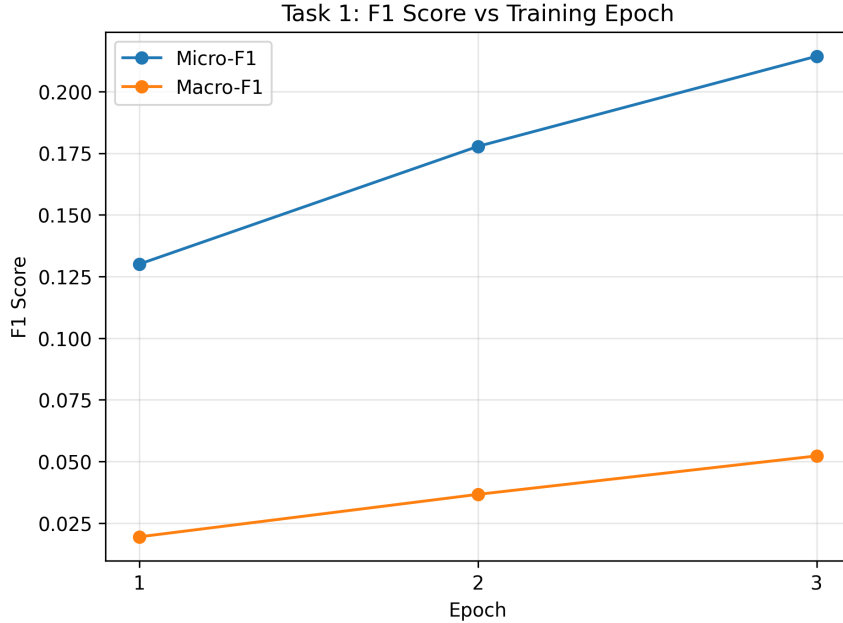


Figure 2: Validation Micro-F1 and Macro-F1 across the three DistilBERT training epochs. These epoch-wise scores use the default decision threshold of 0.5. The final test evaluation uses the validation-selected threshold of 0.20.

5.2 Task 2: GraphSAGE and CNN Comparison

We compare GraphSAGE and the mel-spectrogram CNN on the same 100-track GTZAN test set. The CNN correctly classifies 71 tracks, while GraphSAGE correctly classifies 36. The CNN achieves a Macro-F1 of 0.7112 compared with 0.3258 for GraphSAGE. The same ranking appears on validation data, where the CNN reaches 0.7878 and GraphSAGE reaches 0.3473 Macro-F1.

Performance also varies by genre. GraphSAGE performs relatively well for jazz and classical but fails on reggae, while the CNN performs strongly for classical, metal, and reggae. Rock remains difficult for both models.

A likely reason is the input representation. GraphSAGE uses only six averaged chroma vectors per track, while the CNN receives a richer mel-spectrogram containing more timing and timbre information. Because both the features and model architectures differ, this result does not show that graph learning is generally worse; it only shows that our simple chroma-based graph model is weaker than the implemented CNN on GTZAN.

5.3 Task 3: Multimodal Fusion Comparison

For MTAT, we compare the GNN-only model, BERT-only model, early concatenation, frozen cross-attention, and partial end-to-end cross-attention on the same test split.

Table 3: MTAT test results for the unimodal and fusion models. All models use the same 50 target tags. Bold values show the best result for each metric.

Model	Threshold	Macro-F1	Micro-F1	mAP
GNN-only	0.53	0.1480	0.1924	0.1092
BERT-only	0.74	0.3533	0.3745	0.2941
Early concatenation (frozen)	0.77	0.3654	0.3937	0.3294
Cross-attention (frozen)	0.78	0.3627	0.3866	0.3210
Cross-attention (partial end-to-end)	0.78	0.3615	0.3917	0.3266

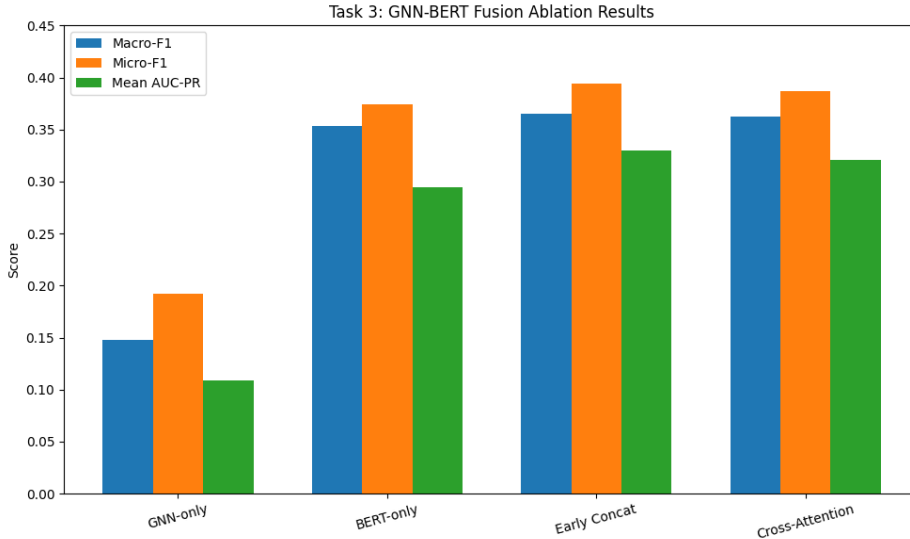


Figure 3: Comparison of the MTAT models using Macro-F1, Micro-F1, and mAP. Early concatenation gives the highest recorded score for all three metrics.

The BERT-only model is much stronger than the GNN-only model. BERT-only achieves a Macro-F1 of 0.3533, while GNN-only reaches 0.1480. This shows that the auxiliary text contains strong information for predicting the music tags.

Adding graph-based audio information improves the results. Early concatenation gives the best overall performance, with a Macro-F1 of 0.3654, Micro-F1 of 0.3937, and mAP of 0.3294. Compared with BERT-only, this improves Macro-F1 by 0.0121, Micro-F1 by 0.0192, and mAP by 0.0353.

Early concatenation also gives the highest selected validation mAP of 0.3518. Frozen cross-attention reaches 0.3421, while partial end-to-end cross-attention reaches 0.3469.

Cross-attention performs close to early concatenation, but it does not improve the final results. Partial end-to-end training improves test mAP from 0.3210 to 0.3266 compared with frozen cross-attention, while its Macro-F1 decreases slightly from 0.3627 to 0.3615.

These results suggest that combining audio and text is useful, but a more complex fusion method does not automatically give better performance. In our experiments, simple early concatenation gives the strongest overall result.

5.4 Qualitative Case Studies

We examine three MTAT examples to understand when the fusion model works well and when it makes mistakes. The selected examples represent a strong prediction, a mixed prediction, and a challenging prediction.

Table 4: Three qualitative MTAT examples. TP, FP, and FN represent true positives, false positives, and false negatives.

Clip / case	Auxiliary text	TP / FP / FN	Jaccard
7944 / strong	folk, man singing	8 / 4 / 1	0.615
2408 / mixed	singer, irish, female singer	3 / 7 / 0	0.300
119 / challenging	no piano	0 / 3 / 2	0.000

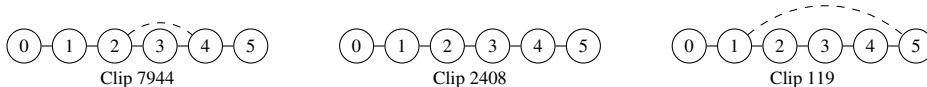


Figure 4: Graph structures for the three qualitative examples. Solid edges connect consecutive audio segments, while dashed edges represent additional similarity connections.

The strong example, with the auxiliary text “folk, man singing,” achieves the highest Jaccard score of 0.615 and correctly predicts several related vocal tags. The mixed example correctly identifies the female-vocal targets but also produces several extra predictions. The challenging example contains only “no piano,” and the predicted tags do not match the annotated *slow* and *flute* labels.

These examples show that useful auxiliary text can support music-tag prediction, while limited or ambiguous text can make prediction more difficult. Figure 4 shows how the six audio segments are connected. The graph paths illustrate the input structure and should not be interpreted as causal explanations of the model predictions.

5.5 Visualization of the Fused Representation

To examine the learned multimodal representation, we visualize 2,000 MTAT test examples using t-SNE. The 256-dimensional frozen cross-attention representations are first reduced to 50 dimensions using PCA, followed by t-SNE with perplexity 30 and 1,000 iterations.

Figure 5 shows the same examples colored using genre-related tags and mood/context-related tags.

The t-SNE plots show that some tags form visible local groups. For example, clips associated with *classical*, *rock*, and *techno* appear more concentrated in some regions of the representation space. However, many examples still remain mixed in the central area.

This overlap is expected because MTAT is a multi-label dataset, so one music clip can have several characteristics at the same time. The mood/context view also includes tags such as *slow*, *fast*, *quiet*,

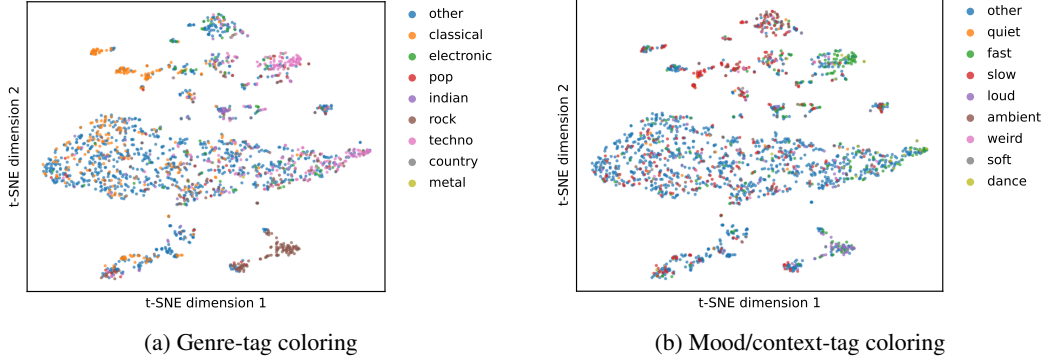


Figure 5: t-SNE visualization of the MTAT fused representations. The same examples are shown in both plots. In (a), the points are colored using genre-related tags. In (b), the same points are colored using mood/context-related tags.

loud, *ambient*, and *dance*. These should be understood as descriptive music-context tags, not as a strict emotion classification system.

Overall, the visualization suggests that the fused model learns some meaningful structure in the music-context representation space, although the labels are not completely separated.

6 Discussion

The experiments show that different representations are useful for different music tasks. On GTZAN, the mel-spectrogram CNN performs much better than the chroma-based GraphSAGE model. One likely reason is that the CNN receives richer time-frequency information, while our graph uses only six averaged chroma vectors.

On MTAT, combining audio and text improves performance compared with either representation alone. Early concatenation gives the best overall result, while cross-attention and partial end-to-end training remain close. This suggests that multimodal information is useful, but a more complex fusion method does not necessarily perform better.

However, we cannot conclude that the graph connections themselves cause the improvement because we did not evaluate a version without graph edges or with simple pooled chroma features. A graph ablation would help separate the contribution of the audio features from the contribution of the graph structure.

The MTAT text input also has an important limitation. The auxiliary text and target labels come from the same annotation vocabulary. Although the exact 50 target columns are removed, related words can still appear, such as “violins” and “violin.” Therefore, we describe the experiment as metadata-assisted music tagging.

The experiments use fixed splits and seed 42, but they were not repeated with several random seeds, and artist-disjointness was not independently verified. Future work should use richer audio features, stronger graph ablations, independent text descriptions, artist-disjoint splits, and repeated runs for more reliable evaluation.

7 Conclusion

This project studies music-context understanding using text, audio graphs, and multimodal fusion across MusicCaps, GTZAN, and MTAT. DistilBERT is used for text-based tagging, while GraphSAGE represents relationships between audio segments. On GTZAN, the mel-spectrogram CNN performs better than the simple chroma-based graph model.

On MTAT, combining GraphSAGE and DistilBERT improves over the unimodal models. Early concatenation gives the strongest result, achieving 0.3654 Macro-F1, 0.3937 Micro-F1, and 0.3294

mAP. Cross-attention and partial end-to-end training perform similarly but do not outperform early concatenation.

Overall, the results show that combining audio and text can improve music-context prediction, while fusion complexity alone does not guarantee better performance. Future work can improve the system using richer audio representations and more controlled evaluation.

References

- [1] Victor Sanh, Lysandre Debut, Julien Chaumond, and Thomas Wolf. DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter. *arXiv preprint arXiv:1910.01108*, 2019. <https://arxiv.org/abs/1910.01108>.
- [2] Will Hamilton, Zhitao Ying, and Jure Leskovec. Inductive representation learning on large graphs. In *Advances in Neural Information Processing Systems*, volume 30, pages 1024–1034, 2017.
- [3] George Tzanetakis and Perry Cook. Musical genre classification of audio signals. *IEEE Transactions on Speech and Audio Processing*, 10(5):293–302, 2002. doi:10.1109/TSA.2002.800560.
- [4] Andrea Agostinelli et al. MusicLM: Generating music from text. *arXiv preprint arXiv:2301.11325*, 2023. <https://arxiv.org/abs/2301.11325>.
- [5] Edith Law, Kris West, Michael Mandel, Mert Bay, and J. Stephen Downie. Evaluation of algorithms using games: The case of music tagging. In *Proceedings of the 10th International Society for Music Information Retrieval Conference*, pages 387–392, 2009.